

Project 2
DevOps Automation Documentation
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Retail Sales Performance Dashboard

1. Project Overview

The Retail Sales Performance Dashboard project was developed to analyze and monitor retail business performance using multiple technologies including MySQL, MongoDB, Python Analytics, PySpark, and Azure Databricks.

The purpose of the DevOps Automation phase is to automate the execution of the retail data pipeline, monitor weekly sales performance, generate reports, and maintain consistent data processing workflows.

This documentation explains the proposed Azure DevOps automation workflow for the project.

2. Objective of DevOps Automation

The main objectives of the DevOps automation pipeline are:

- Automate weekly retail sales processing
- Reduce manual execution effort
- Generate scheduled business reports
- Monitor underperforming products and stores
- Maintain reliable ETL execution
- Improve data consistency and operational efficiency

3. Technologies Used

Technology	Purpose
Azure DevOps	CI/CD and workflow automation
Azure Databricks	ETL pipeline execution
PySpark	Big data processing
Python	Analytics and reporting
MySQL	Structured retail sales storage
MongoDB Atlas	Customer campaign feedback storage
Google Colab	Python and PySpark execution

4. Folder Structure

```
Retail_sales_folder/  
├── mysql/
```

```
|— mongodb/  
|— python_analytics/  
|— pyspark/  
|— databricks/  
|— azure_devops/  
|— dataset/
```

5. Input Data

The pipeline uses four retail datasets.

5.1 products.csv

Contains:

- Product details
- Category information
- Product pricing
- Product cost
- Brand information

Sample Fields

Column Name
product_id
product_name
category
price
cost
brand

5.2 stores.csv

Contains:

- Store details
- Regional information
- Manager information

Sample Fields

Column Name
store_id
store_name

city
region
manager_name

5.3 employees.csv

Contains:

- Employee information
- Store mapping
- Salary details

Sample Fields

Column Name
employee_id
employee_name
designation
store_id
salary

5.4 sales.csv

Contains:

- Retail sales transactions
- Product sales details
- Quantity sold
- Discounts and payment modes

Sample Fields

Column Name
sale_id
sale_date
store_id
product_id
employee_id
quantity
discount_percent
payment_mode

6. Proposed DevOps Workflow

The DevOps workflow automates the complete retail data processing cycle.

Workflow Stages

- 1. Dataset upload
- 2. Data validation
- 3. Bronze layer ingestion
- 4. Silver layer cleaning
- 5. Gold layer analytics
- 6. KPI generation
- 7. Weekly report generation
- 8. Output storage
- 9. Monitoring and logging

7. Data Processing Architecture



8. Pipeline Scheduling

The DevOps pipeline is configured to run automatically every Monday.

Schedule

Parameter	Value
Frequency	Weekly
Day	Monday
Time	09:00 AM

Purpose:

- Weekly retail sales analysis
- Weekly KPI generation
- Automated monitoring

9. Bronze Layer Processing

The Bronze layer is responsible for:

- Raw dataset ingestion
- Reading CSV files
- Initial storage
- Preserving source data

Bronze Tables

Table
bronze_sales
bronze_products
bronze_stores
bronze_employees

10. Silver Layer Processing

The Silver layer performs:

- Data cleaning
- Null value removal
- Data standardization
- Data transformation
- Revenue calculations

Silver Tables

Table
silver_sales
silver_products
silver_stores
silver_employees
silver_retail_analytics

Processing Activities

- Uppercase conversion
- Revenue calculation
- Discount calculation
- Profit calculation
- Dataset joins
- Duplicate handling

11. Gold Layer Processing

The Gold layer generates business insights.

Gold Tables

Table
gold_store_kpis
gold_top_products
gold_underperforming_products
gold_region_sales_report

12. KPI Calculations

The pipeline calculates multiple business KPIs.

Store KPIs

- Total revenue
- Total profit
- Average discount percentage
- Total sales count

Product KPIs

- Top selling products
- Underperforming products
- Revenue by product

Regional KPIs

- Regional revenue
- Regional profit

13. Revenue Calculation Logic

Revenue Formula

Revenue = Quantity × Product Price

Discount Amount Formula

Discount Amount = Revenue × Discount Percentage / 100

Final Revenue Formula

Final Revenue = Revenue – Discount Amount

Profit Formula

Profit = Final Revenue – Product Cost

14. Underperforming Product Detection

Products are considered underperforming when:

- Quantity sold is less than 3
- Revenue is low
- Sales frequency is low

Purpose

- Improve inventory management
- Identify low-performing items
- Improve sales strategy

15. Output Files Generated

The automated pipeline generates the following outputs.

Output File	Purpose
cleaned_sales.csv	Cleaned retail dataset
summary_report.csv	Store revenue summary
underperforming_products.csv	Low sales products report
store_summary.csv	Store-level KPI report

16. Monitoring and Logging

The DevOps pipeline maintains execution logs.

Logged Information

- Pipeline execution status
- ETL processing status
- Data loading success/failure
- Transformation completion
- Output generation status

17. Error Handling

The proposed pipeline handles:

- Missing values
- Duplicate columns
- Schema mismatches
- Invalid records
- Null records

Example

Duplicate column conflicts during Databricks joins were resolved by removing repeated columns before joining datasets.

18. Security and Access Control

The system uses:

- Azure role-based access
- Controlled dataset access
- Cloud-based storage
- Secure pipeline execution

19. Benefits of DevOps Automation

Benefit	Description
Automation	Reduces manual execution
Reliability	Consistent workflow execution
Scalability	Handles growing datasets
Monitoring	Tracks execution status

Reporting	Generates business insights automatically
Efficiency	Improves operational performance

20. Expected Output

After successful execution, the system provides:

- Automated weekly ETL processing
- Business KPI dashboards
- Store-level revenue analysis
- Underperforming product reports
- Regional sales reports
- Data visualization outputs

21. Conclusion

The Retail Sales Performance Dashboard successfully demonstrates the integration of SQL, NoSQL, Python Analytics, Big Data Processing, and Cloud ETL concepts.

The proposed DevOps automation workflow improves efficiency by automating weekly retail sales processing and KPI generation. The pipeline supports scalable retail analytics and provides valuable business insights through Bronze, Silver, and Gold layer processing.

The project demonstrates practical implementation of modern Data Engineering and DevOps concepts using industry-oriented tools and workflows.